Ensemble Learning

Bagging, boosting, stacking, etc.

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Definição de “ensemble learning”

- Se você agregar as previsões de um conjunto de previsores (como classificadores ou regressores), muitas vezes obterá melhores previsões do que com o melhor previsor individual. Um conjunto de previsores é chamado de ensemble; assim, esta técnica é chamada Ensemble Learning, e um algoritmo de Ensemble Learning é chamado de Ensemble method.
- Exemplo: Random forest, uma “ensemble” de árvores de decisão

Voting classifiers

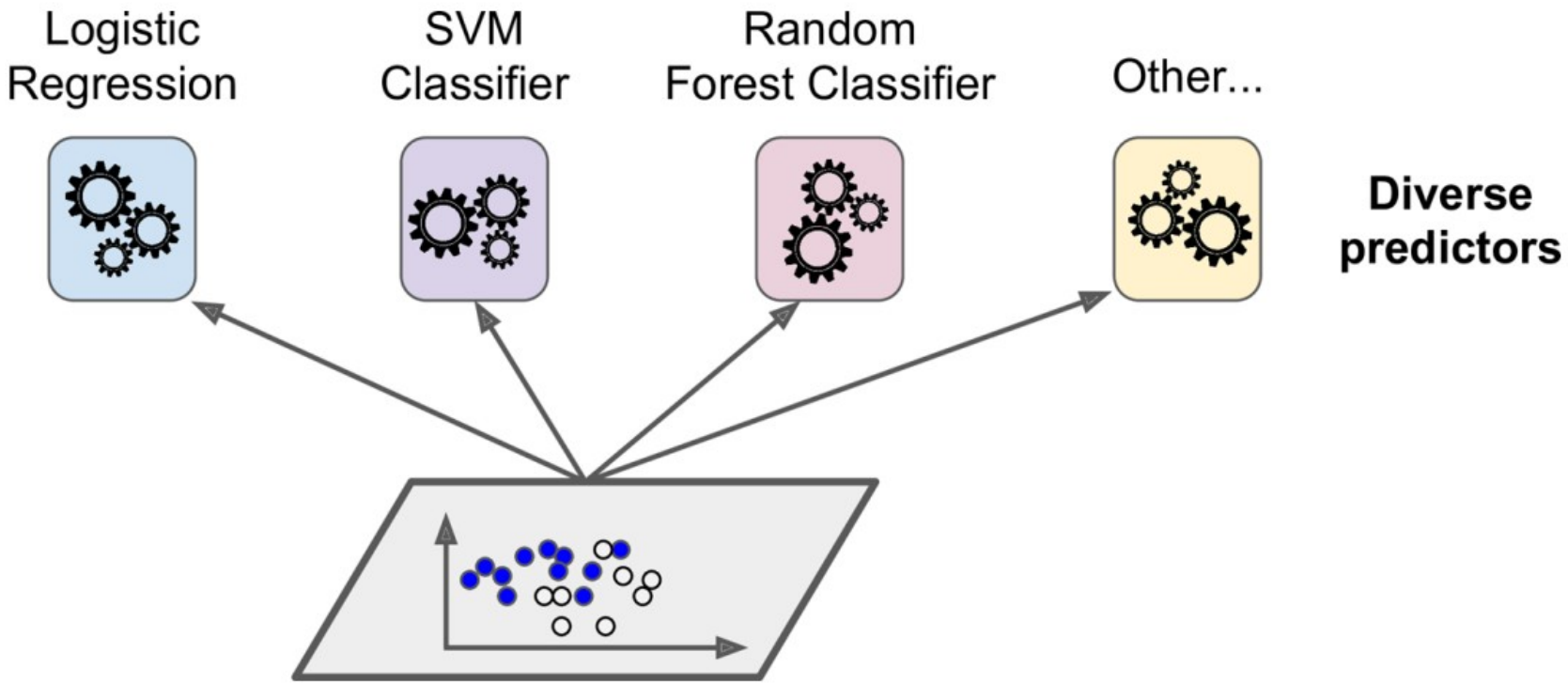

Figure 7-1. Training diverse classifiers

Hard voting

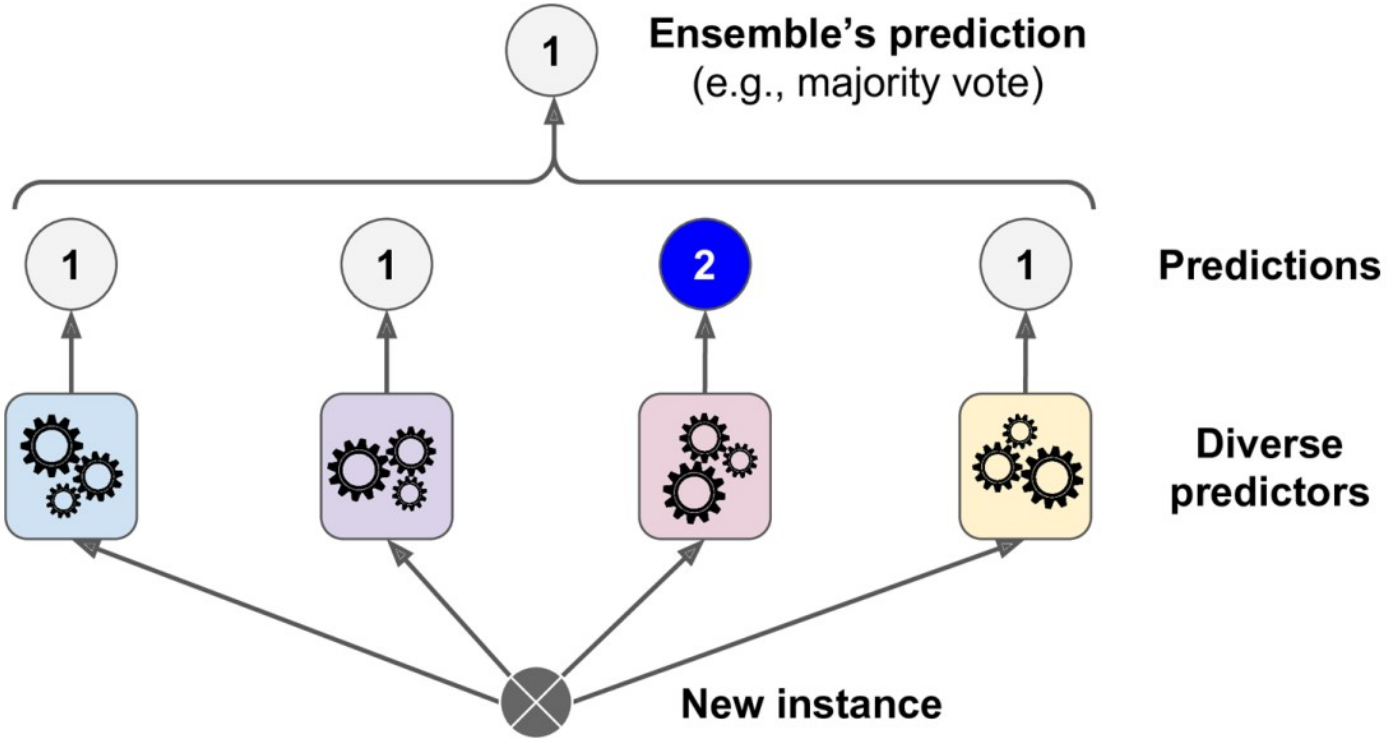

Figure 7-2. Hard voting classifier predictions

Preditores devem ser “independentes”

TIP

Ensemble methods work best when the predictors are as independent from one another as possible. One way to get diverse classifiers is to train them using very different algorithms. This increases the chance that they will make very different types of errors, improving the ensemble’s accuracy.

Exemplo de código

```
from sklearn.ensemble import RandomForestClassifier
from sklearn.ensemble import VotingClassifier
from sklearn.linear_model import LogisticRegression
from sklearn.svm import SVC

log_clf = LogisticRegression()
rnd_clf = RandomForestClassifier()
svm_clf = SVC()

voting_clf = VotingClassifier(
    estimators=[('lr', log_clf), ('rf', rnd_clf), ('svc', svm_clf)],
    voting='hard')
voting_clf.fit(X_train, y_train)
```

Soft voting

If all classifiers are able to estimate class probabilities (i.e., they all have a `predict_proba()` method), then you can tell Scikit-Learn to predict the class with the highest class probability, averaged over all the individual classifiers. This is called *soft voting*. It often achieves higher performance than hard voting because it gives more weight to highly confident votes. All you need to do is replace `voting="hard"` with `voting="soft"` and ensure that all classifiers can estimate class probabilities. This is not the case for the `SVC` class by default, so you need to set its `probability` hyperparameter to `True` (this will make the `SVC` class use cross-validation to estimate class probabilities, slowing down training, and it will add a `predict_proba()` method).

Bagging and Pasting

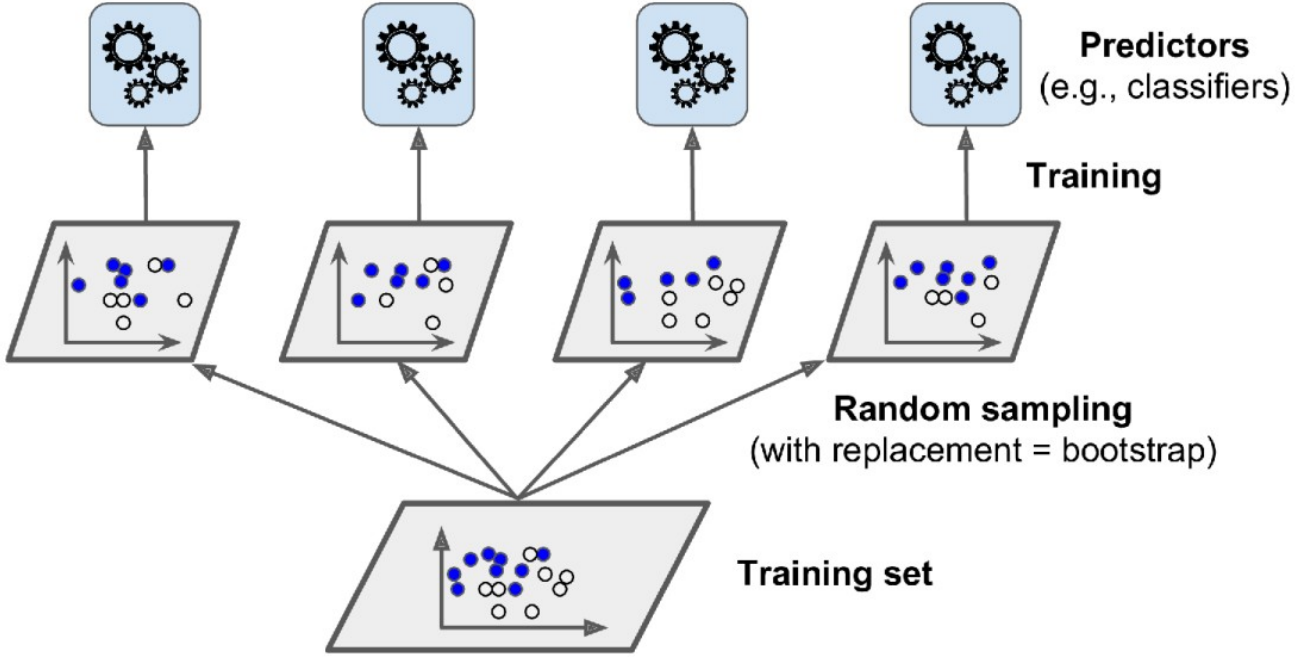

Figure 7-4. Bagging and pasting involves training several predictors on different random samples of the training set

Example

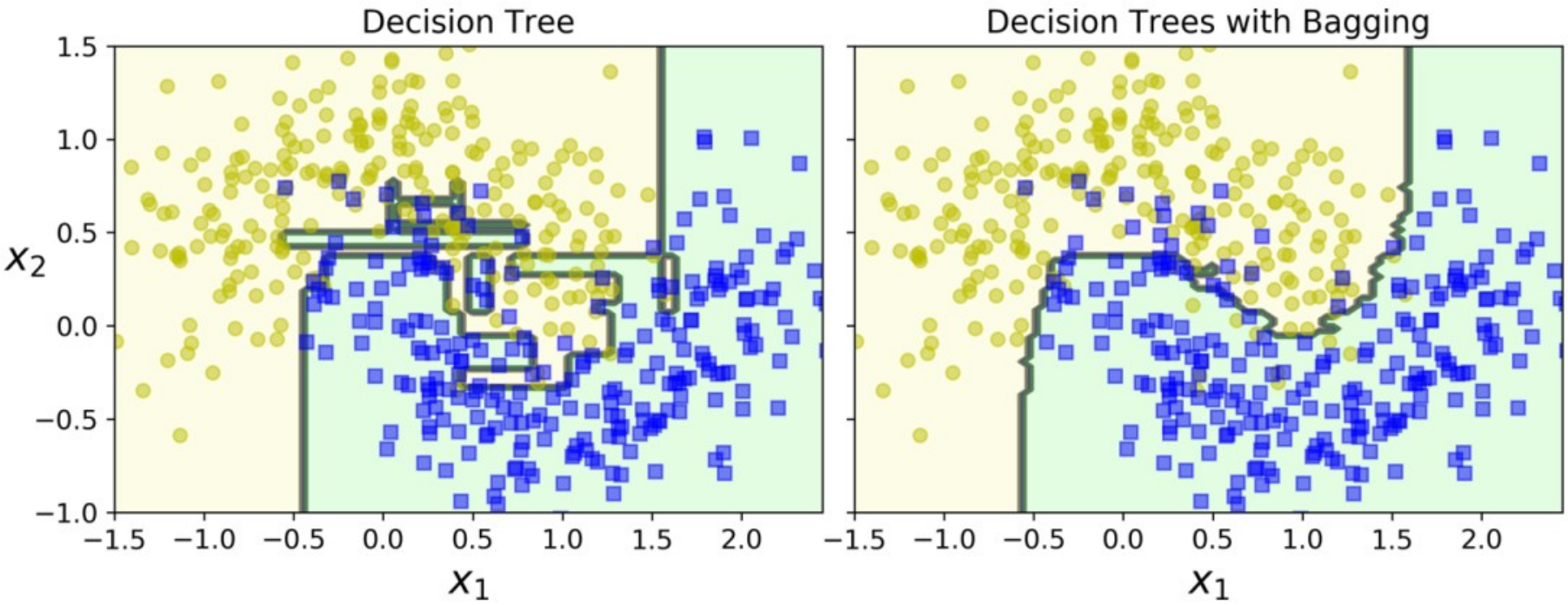

Figure 7-5. A single Decision Tree (left) versus a bagging ensemble of 500 trees (right)

Stacking

- instead of using trivial functions (such as hard voting) to aggregate the predictions of all predictors in an ensemble, why don’t we train a model to perform this aggregation

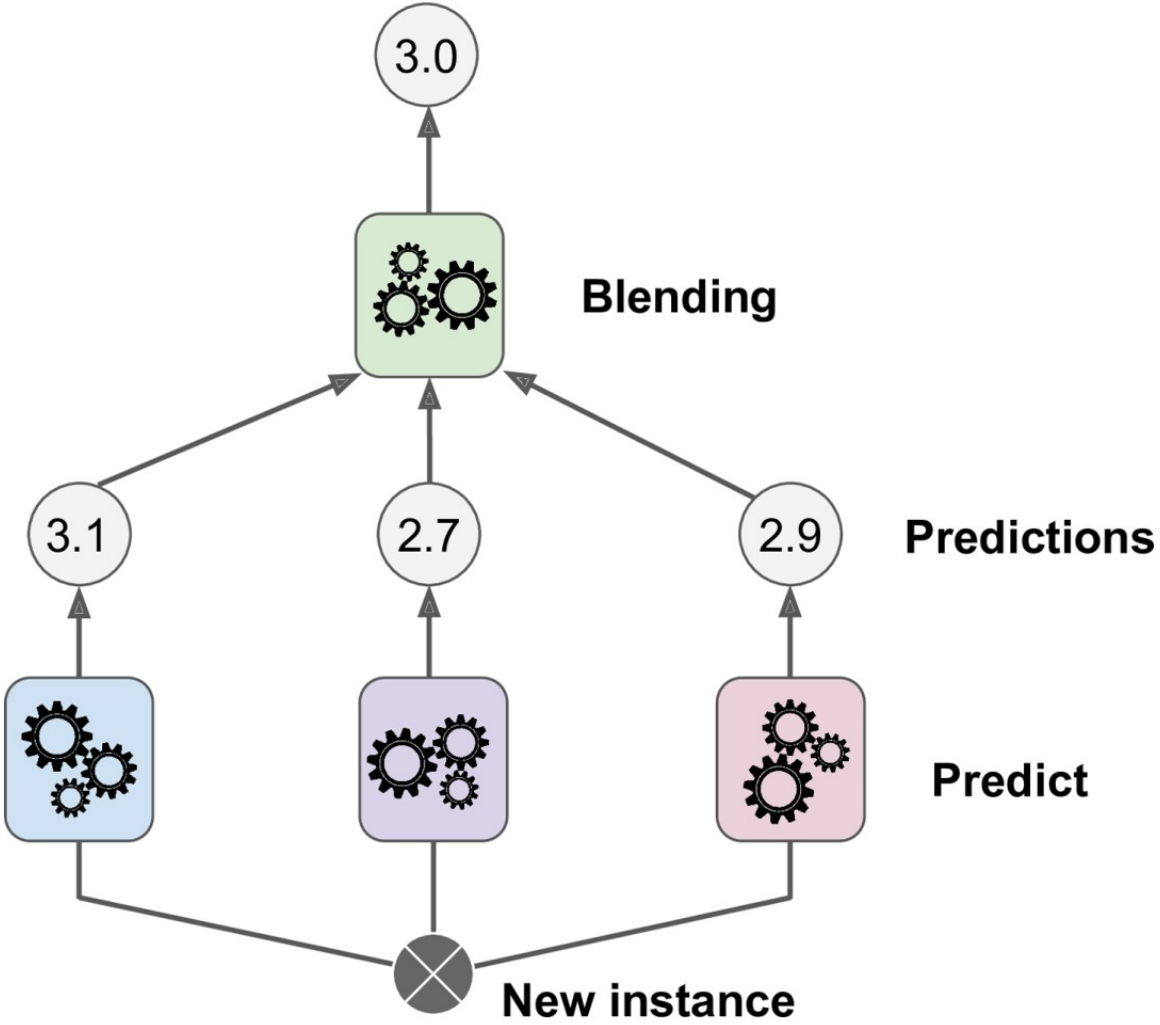

Figure 7-12. Aggregating predictions using a blending predictor